

# Gloria LMS: analysis, requirements and MVP definition

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Gloria's course delivery process relied on Google Drive, Excel and Microsoft Teams. The project examined where this setup failed, used the findings to define product requirements and reduced the planned scope to an MVP that users could test.

# Understanding the operational problem

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The work began with Gloria's operating model and the recurring problems reported by the Course Manager. The team first established which course activities belonged within the LMS scope.

**The first question concerned the product boundary**

# Fragmented operations made course status unreliable

Evidence

PM interview  
Company context  
Existing tools

Drive, Excel and Teams supported separate parts of the course lifecycle. Staff reconstructed the overall status manually.

Where the process failed

- 01 Course materials could disappear or lose ownership
- 02 Teams rebuilt course status from several sources
- 03 Manual coordination made deadlines easy to miss

Business requirements

- BR-01 Protect course materials and operational knowledge
- BR-03 Reduce manual coordination and missed actions
- BR-04 Monitor each course using reliable data

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**The business problem did not yet define which course activities belonged within the LMS scope.**

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# The LMS scope covers course operations for each course edition

The boundary separates course operations from commercial, financial and live-delivery systems.

## Within the LMS scope

**Course edition and participants**  
**Schedule and changes**  
**Materials and assignments**  
**Submissions, feedback and progress**  
**Operational status and basic reporting**

## Connected services

CRM provides learner and client context  
Payments provide settlement status  
Teams or Zoom deliver live sessions  
Email and messaging deliver notifications

## Outside the MVP

Sales pipeline and marketing automation  
Payments, invoicing and accounting  
Full helpdesk and contact centre  
Strategic BI and lecturer contracts

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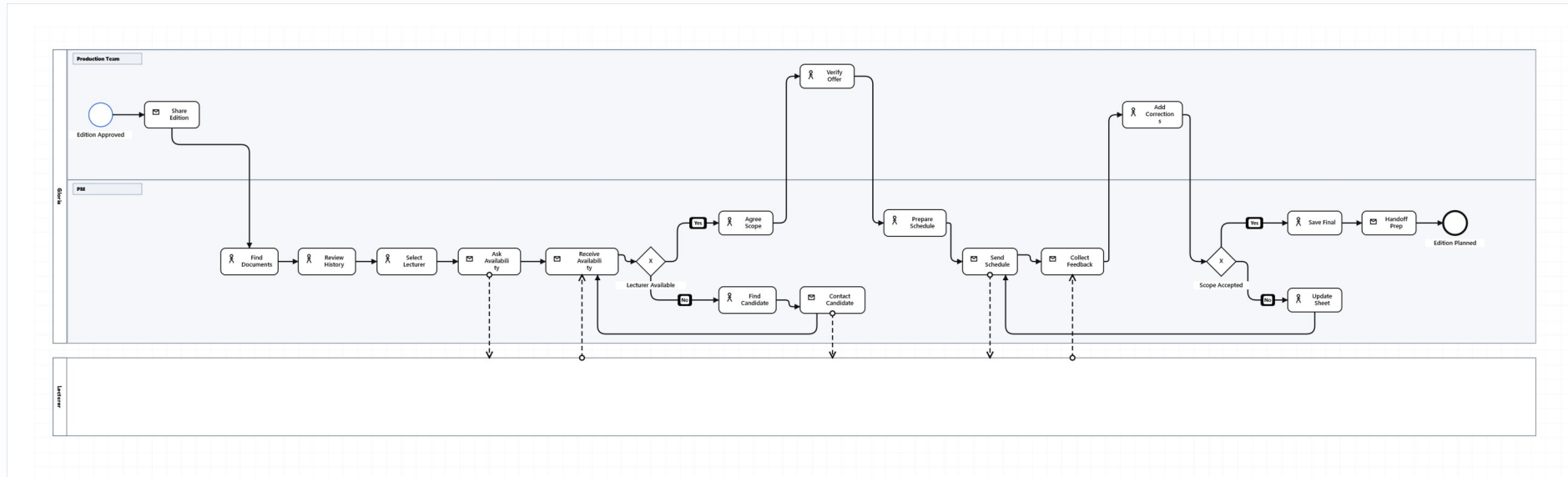
**After setting the boundary, the team mapped the current process to identify where the same problems occurred repeatedly.**

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# The same coordination problems occurred at several process stages

## REPRESENTATIVE ARTIFACT

The team mapped course planning end to end to check whether the reported tool problems affected one task or several process stages.



Findings from the map

Manual handoffs

Status rebuilt across tools

No single operational record

## EVIDENCE

The same failure pattern appeared across several lifecycle handoffs

# The AS-IS analysis identified recurring operational problems

## Interview signals

### Drive

Files and ownership were distributed

### Excel

Status depended on manual updates

### Manual coordination

Teams reconciled changes across tools

## AS-IS processes

### Course planning

### Course preparation

### Course administration

### Reporting

The same issues appeared at several handoffs instead of one isolated task.

## Recurring failure points

- 01 No shared ownership of course information
- 02 Teams synchronized status manually
- 03 No current source of course information

FINDING

The interview described the symptoms. AS-IS analysis identified where they recurred

# From evidence to requirements

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The interview first suggested a problem with tools. Process analysis showed that teams lacked shared responsibility for course information and a reliable current status. The team recorded both the user need and the strength of the supporting evidence.

**The diagnosis changed the question from tools to product responsibility**

# User requirements reflected different levels of evidence

The PM perspective was direct. Learner and Lecturer needs remained working hypotheses until later validation.

## COURSE MANAGER

### DIRECT EVIDENCE

- See the current course status
- Control deadlines and changes
- Monitor without rebuilding reports

## LEARNER

### REPORTED BY PM

- Find tasks and deadlines quickly
- Access materials in one place
- Submit work and receive feedback

## LECTURER

### HYPOTHESIS TO VALIDATE

- Publish current materials
- Review submissions efficiently
- See what requires a response

### EVIDENCE GAP

**Later research must confirm the needs inferred for Learners and Lecturers**

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**Rather than combine all needs into one feature list, the team recorded the source, evidence status and purpose of each requirement.**

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# The team developed requirements from business problems and user needs

| Problem          | Business requirement       | User requirement      | Functional requirement         | Quality constraint          |
|------------------|----------------------------|-----------------------|--------------------------------|-----------------------------|
| Lost materials   | Prevent knowledge loss     | One trusted place     | Repository and versioning      | Data integrity and access   |
| Missed deadlines | Reduce operational risk    | Course overview       | Schedule, status and reminders | Usability                   |
| Manual reporting | Support reliable decisions | Monitor without Excel | Reporting and monitoring       | Performance and consistency |

TRACEABILITY

Each requirement links a business problem to system behaviour and test conditions

# Each catalogue entry records its source and validation status

CATALOGUE EXCERPT

Three representative chains show how business intent remained connected to user needs, system behaviour and quality constraints.

| Problem                             | BR                               | UR                                | FR                                        | NFR                                   | Source                 | Status             |
|-------------------------------------|----------------------------------|-----------------------------------|-------------------------------------------|---------------------------------------|------------------------|--------------------|
| Materials and knowledge can be lost | BR-01<br>Prevent knowledge loss  | UR-PM-01<br>One trusted place     | FR-04/05<br>Publish and version materials | NFR-REL-01<br>Data integrity          | PM interview + AS-IS   | Direct / validated |
| Deadlines and status are missed     | BR-03<br>Reduce operational risk | UR-PM-02<br>Course overview       | FR-02/10<br>Schedule and monitor          | NFR-US-01<br>Clear overview           | PM interview + process | Direct / derived   |
| Reporting depends on Excel          | BR-04<br>Reliable monitoring     | UR-PM-04<br>Monitor without Excel | FR-10/12<br>Monitoring and reporting      | NFR-PERF-01<br>Consistent performance | PM interview           | Direct / validate  |

The full catalogue remains a working document. This excerpt shows how traceability was recorded.

CATALOGUE STRUCTURE

Each entry includes the related problem, source and validation status

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**After defining system behaviour, the team set the quality conditions that would make that behaviour dependable in daily work.**

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# The MVP also needs a clear quality baseline

These constraints address the access, reliability and compatibility problems found in the current process.

## — Usability

One clear access path  
Role-specific navigation

## — Reliability

Stable ownership of course data  
Consistent reporting

## — Security

Role-based access  
Protected learner data

## — Performance

Responsive core workflows  
Groups up to about 90 learners

## — Compatibility

Browser, Android and iOS  
Links to live tools

## — Accessibility

Readable structure  
Keyboard and responsive use

### BASELINE

**Core workflows must remain understandable, secure and reliable across devices**

# Defining the product response

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The requirements baseline now covered business goals, user needs, system behaviour and quality constraints. The next task was to combine these layers into one coherent operating model for the product.

**The requirements define the obligations. The target model shows how roles, actions and data connect**

# The product model connects user needs with system requirements

User need

COURSE MANAGER

A trusted view of course status and exceptions

LEARNER

One place for tasks, deadlines and feedback

LECTURER

A clear way to publish, assess and respond

TO-BE capabilities

Course edition

Submission and feedback

Schedule

Operations monitoring

Materials

Access and reporting

Assignments

Shared course records

Functional requirements

FR-02      Manage schedule

FR-04      Publish materials

FR-07      Submit assignment

FR-08      Grade and provide feedback

FR-10      Monitor course operations

FR-15      Manage access

# Three roles operate the same course lifecycle

The use case model assigns actions to each role while all roles use the same course information.

## LECTURER

- Publish materials
- Create assignments
- Review submissions
- Grade and give feedback

## LEARNER

- View materials
- Open assignment details
- Submit work
- View grades and feedback

## COURSE MANAGER

- Create course edition
- Manage schedule and participants
- Monitor operations
- View reporting and handle issues

Shared access

Course communication

Course status

Exceptions and reporting

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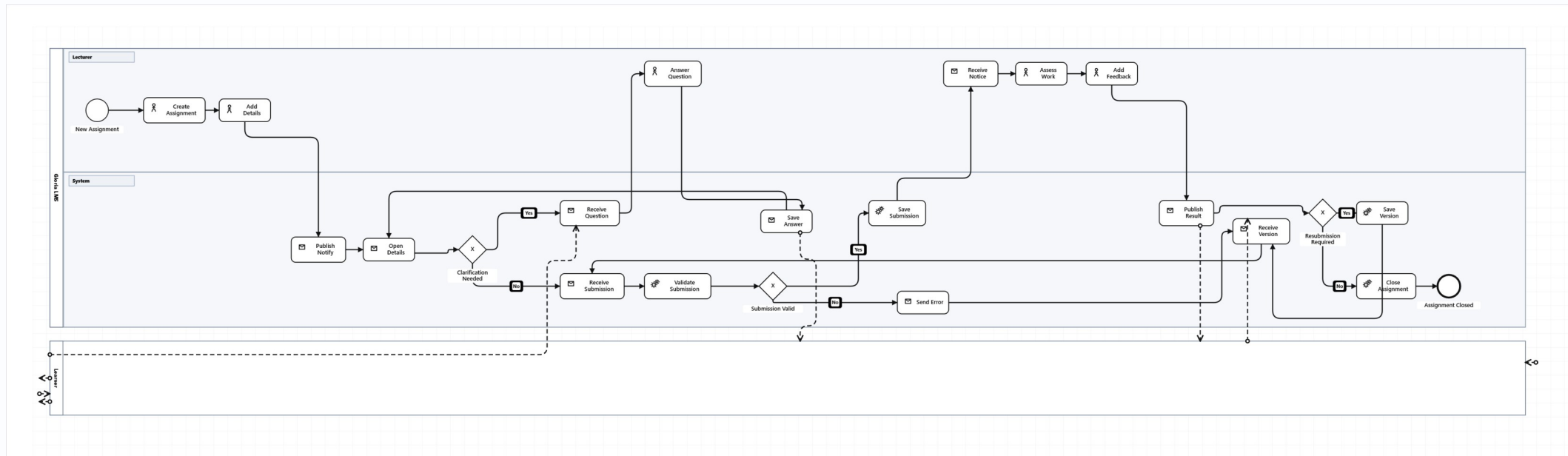
**The use case model assigned actions to each role. The team then used the TO-BE process to specify how each action changes the recorded course status.**

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# The TO-BE process identifies changes in course status recorded by the LMS

## REPRESENTATIVE ARTIFACT

The assignment and feedback flow shows how user actions, system checks and shared status replace manual reconciliation.



State recorded by LMS

Published assignment

Validated submission

Feedback and result

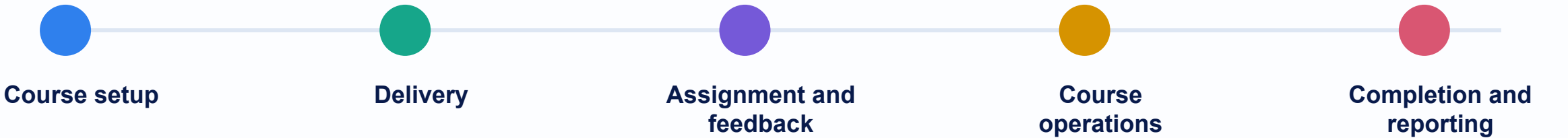
Closed or reopened

## PROCESS CHANGE

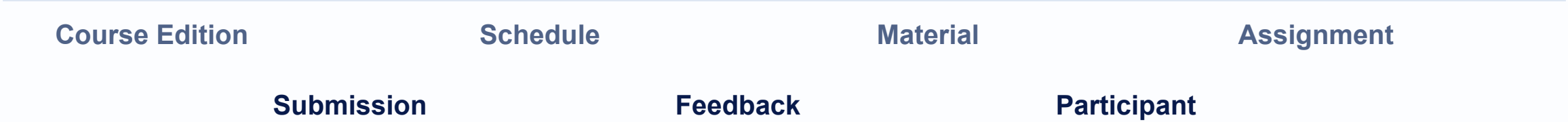
The LMS records the course status after each user action

# Course information and processes included in the LMS scope

## Course lifecycle



## Shared course records



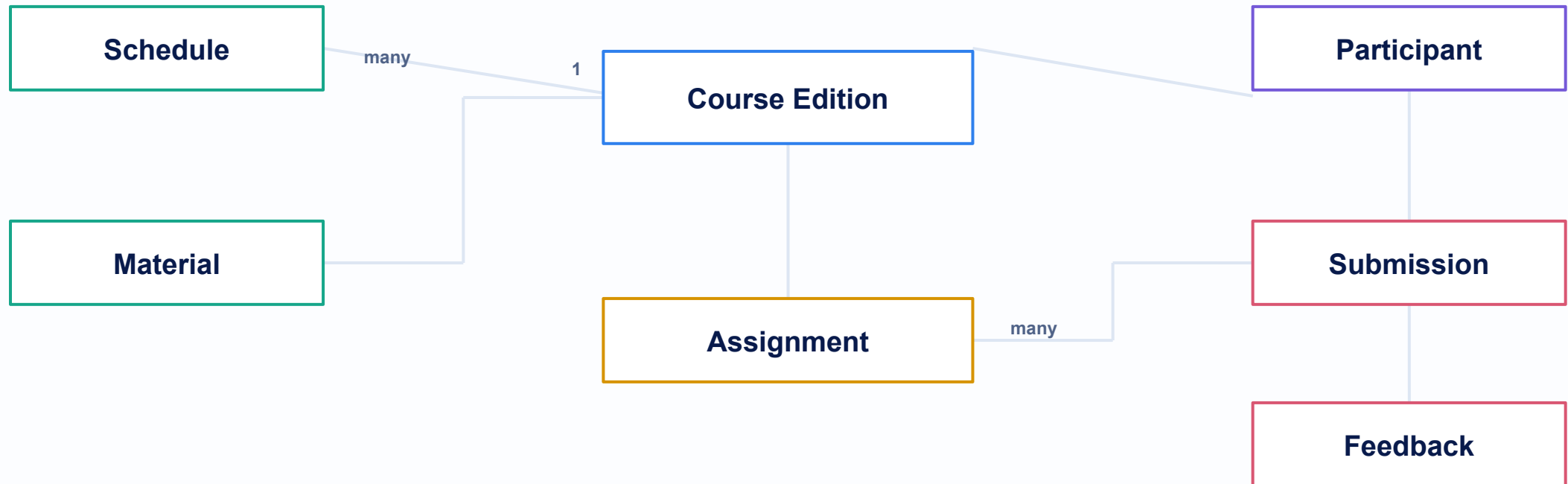
The model separates delivery, course operations and exception handling while using the same course records.

LMS RESPONSIBILITY

Store the current course status across role changes and process handoffs

# Domain model for shared course information

Course Edition is the main record that connects the schedule, materials, participants, assignments, submissions and feedback.



# Reducing the solution to a testable MVP

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Together, the requirements, TO-BE process and domain model described how the LMS stores current course information. The team then had to select the smallest complete workflow involving all three roles.

**The first release had to cover a complete workflow that users could test**

# Assignment lifecycle met the criteria for the first MVP test

The comparison covered business value, roles involved, testability and implementation dependencies.

| Candidate slice     | Business value | Roles covered | Testability | Dependencies | Decision              |
|---------------------|----------------|---------------|-------------|--------------|-----------------------|
| Course setup        | High           | Medium        | Medium      | High         | Later                 |
| Materials           | High           | Medium        | High        | Low          | Supporting capability |
| Assignment workflow | High           | High          | High        | Medium       | Selected for the MVP  |
| Reporting           | Medium         | Low           | Medium      | High         | Later                 |

The selected slice combines materials, deadlines, submission, feedback and course status monitoring.

DECISION

The selected slice covers a complete workflow involving all three roles

# The assignment workflow was selected for the first MVP

| Selection criterion            | Evidence within the slice                                                     |
|--------------------------------|-------------------------------------------------------------------------------|
| Involves all three roles       | Lecturer, Learner and Course Manager update the same course record            |
| Uses shared course information | Materials, deadlines, submissions, status and feedback must stay consistent   |
| Has a clear start and end      | Publishing starts the workflow. Course monitoring confirms its completion     |
| Covers key system functions    | The workflow includes delivery, assessment, permissions and status visibility |
| Works without the full LMS     | The team can validate the model before building every capability              |

PublishDiscoverSubmitReviewFeedbackMonitor

SELECTION LOGIC

The selected workflow lets all three roles use and update the same course information

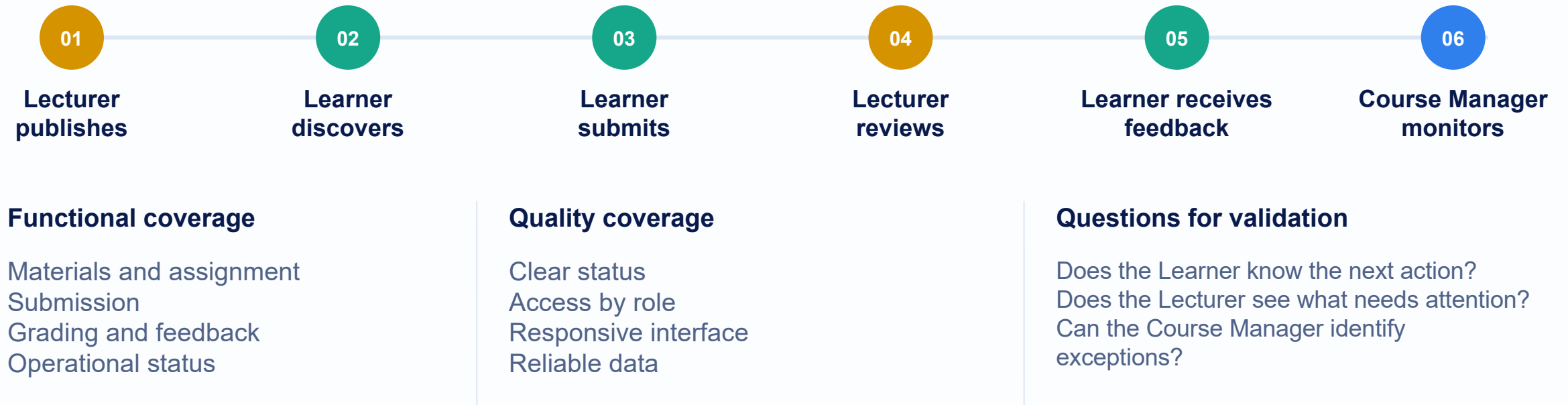
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**After selecting the assignment lifecycle, the team defined the functional and quality scope required to test it end to end.**

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# Scope of the first MVP: one complete assignment workflow

## Assignment lifecycle



# From product definition to UX

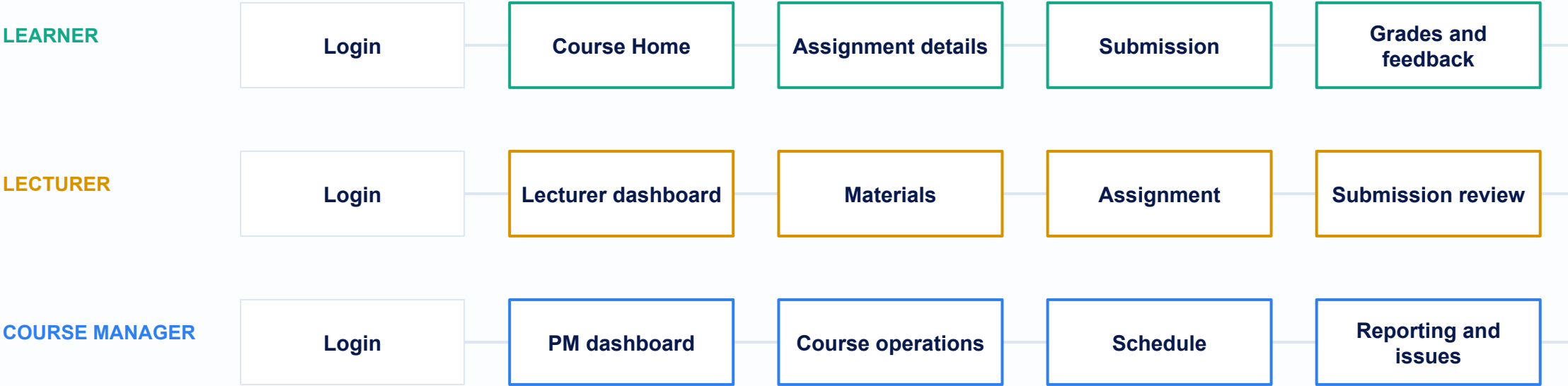
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After defining the MVP scope, the team still needed to test how each role would use it. Each role had to complete the assignment workflow and understand the current course status without coordinating through other tools.

**Next artifacts: user flows, screen designs and prototype validation**

# Core user flows included in the MVP

Each flow ends with a user outcome and updates the same course records.



Planned next artifacts: Information Architecture, Screen Inventory and prototype states.

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**The team can now use these flows to prepare the Information Architecture, Screen Inventory, wireframes and prototype scenarios.**

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# Navigation structure for shared and role-specific pages

Course Overview contains the current course context. Each role sees the actions related to its responsibilities.

|                |                    |                    |                           |                      |
|----------------|--------------------|--------------------|---------------------------|----------------------|
| SHARED         | Dashboard          | My Courses         | Course Overview           | Notifications        |
| LEARNER        | Course Home        | Assignment details | Submission                | Grades and feedback  |
| LECTURER       | Lecturer dashboard | Materials          | Assignment management     | Submission review    |
| COURSE MANAGER | PM dashboard       | Course setup       | Schedule and participants | Reporting and issues |

# Seven screens form the first design and validation set

The selection covers orientation, the assignment workflow and course monitoring without designing the whole LMS.

| Screen                          | Primary role   | Reason to prototype first                           | Flow covered          |
|---------------------------------|----------------|-----------------------------------------------------|-----------------------|
| Dashboard                       | Shared         | Shows current priorities and next actions           | Entry and orientation |
| Course Overview                 | Shared         | Keeps schedule, materials and status in one context | Course navigation     |
| Learner Course Home             | Learner        | Tests whether the next task is clear                | Discover              |
| Assignment Details + Submission | Learner        | Tests brief, deadline and submission confirmation   | Submit                |
| PM Dashboard                    | Course Manager | Tests visibility of risks and exceptions            | Monitor               |
| Schedule Management             | Course Manager | Tests updates to the shared course schedule         | Coordinate            |
| Course Materials Management     | Lecturer       | Tests publishing and ownership of current content   | Publish               |